FDTD: diffraction over objects

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December 2025

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3.1 Results

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4.1 Finite Impedance

4.2 Refinement

In order to better model the non-cartesian surface, we included a grid refinement around the tilted boundaries.

This was achieved by making dx and dy functions of respectively x and y. As a first attempt both were made a step-function, in which around the boundaries they are both half of what they are elsewhere. This however results in reflections at the discontinuity in dx or dy, which can be seen when looking at an animation of a short Gaussian pulse. Next, to remove these reflections, a buffer zone was added in both x- and y-direction in which dx and dy are linearly interpolated (see fig 1).

Both these were in practice implemented by taking the original discretisation coordinates, and transforming these, either partswise linearly (for the step-function in dx or dy), or with a parabolic part in the bufferzone (for the linear dx and dy).

There were dx or dy is not constant however, this results in an error in the update scheme. As, for example, the discretised o_x may no longer be halfway between two p's. This means that in the update equations of o_x, it is erronous to use $\frac{dp}{dx}$ as calculated by differentiating two neighbouring p's. So an interpolation

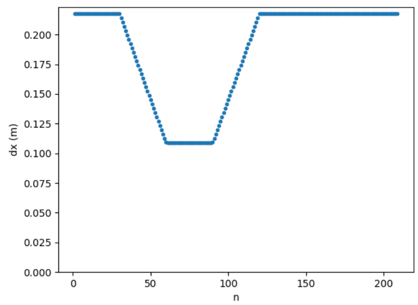

Figure 1: dx after refinement (linear)

was made between three values of $\frac{dp}{dx}$ at the halfway point between two p's: two closest neightbours to the left, the two direct neighbours, and those to the right. Since there are three values, a parabolic interpolation was optimal. At the edges, where there are only two values to interpolate between, a linear interpolation was used instead. To update o_x, the interpolated $\frac{dp}{dx}$ at the location of o_x (after the coordinate transformation) was used.

Some results from the refinement and comparison with without: INSERT HERE.

4.3 Non Cartesian Grid

Another route that was explored is to tilt the grid. In this way the surface of the tilted boundary becomes flat on one side, thus removing the problems that arise from a staircase-like boundary. The left-side was chosen to be flat, as the most important reflections occur there.

Instead of working with x- and y-coordinates, a switch is made to u- and v-coordinates. These are related via equations 1 and 2, in which θ is the angle between the v-axis and the x-axis.

$$x = u + v\cos\theta \tag{1}$$

$$y = v\sin\theta \tag{2}$$

In figure 2 the basis-vectors are given. Note that $\vec{e_x} = \vec{e_u}$, but $x \neq u$. In this figure another vector, $\vec{e_w}$, is introduced. This vector is perpendicular to $\vec{e_v}$. With this figure the relations given in equations 3 and 4 are easily derived.

$$\vec{e_w} = \sin\theta\vec{e_x} - \cos\theta\vec{e_y} \tag{3}$$

$$\vec{e_v} = \cos\theta\vec{e_x} + \sin\theta\vec{e_y} \tag{4}$$

There is still an important choice remaining, namely, which directions of $\vec{o}$ to store. A first idea might be to choose $o_u = \vec{e_u} \cdot \vec{o}$ and $o_v = \vec{e_v} \cdot \vec{o}$, as analogous

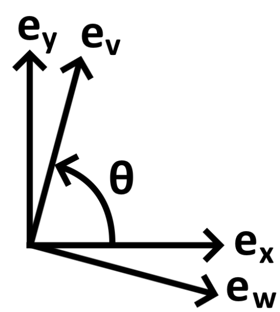

Figure 2: Important unit vectors

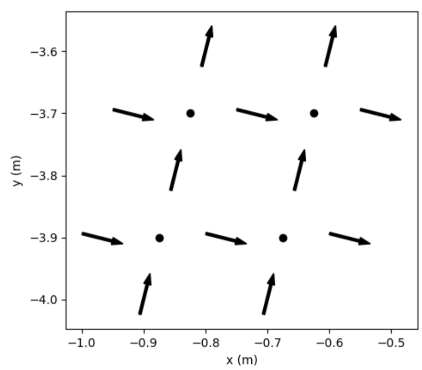

Figure 3: Discretised fields

to the cartesian grid. Remember however that the goal is to implement a tilted PEC-boundary, which has as boundary condition that $o_n = 0$, with n the normal of the boundary. For this boundary condition to be implemented simply, o_w and o_v are chosen, in which $\vec{e_w}$ is the normal of the tilted boundary (on the left side). An added benefit is that now both stored directions of $\vec{o}$ are orthogonal to eachother, so no data overlaps. Figure 3 shows the location of a few p, o_w and o_v's.

4.3.1 Update Scheme

Now the update scheme can be adjusted for the tilted grid. To update p, the divergence of $\vec{o}$ is needed at the location of p, with $\nabla \cdot \vec{o} = \frac{do_w}{dw} + \frac{do_v}{dv}$. The following quantities are known:

$$\frac{do_v}{dv}|_{v_0+\Delta v/2} \cdot \Delta v \approx o_v|_{v_0+\Delta v} - o_v|_{v_0},$$

$$\frac{do_w}{du}|_{u_0+\Delta u/2} \cdot \Delta u \approx o_w|_{u_0+\Delta u} - o_w|_{u_0}.$$

$\frac{do_w}{dw}$ requires some more attention. From equations 4 and 3, the following can be derived:

$$\vec{e_w} = \frac{1}{\sin\theta}\left(\vec{e_u} - \cos\theta\vec{e_v}\right).$$

From which

$$\frac{do_w}{dw} = \vec{e_w}\cdot\nabla o_w = \frac{1}{\sin\theta}\left(\frac{do_w}{du} - \cos\theta\frac{do_w}{dv}\right)$$

follows. So the missing piece is $\frac{do_w}{dv}$, which can only be interpolated at p's location by using

$$\frac{do_w}{dv}|_{v_0}\cdot 2\Delta v \approx o_w|_{v_0+\Delta v} - o_w|_{v_0-\Delta v},$$

combined with

$$o_w|_{u_0+\Delta u/2} \approx \frac{o_w|_{u_0} + o_w|_{u_0+\Delta u}}{2}.$$

For the update equations of $\vec{o}$ an analogous approach is followed.

In a hand-waving approach a rough approximation was made for the Courant Number. Starting from the standard form, and substituting $dx = du + dv\cos\theta$ and $dy = dv\sin\theta$ (derived from equations 1 and 2), equation 5 is calculated.

$$\mathrm{CN} = c^2 dt^2\left((du + dv\cos\theta)^{-2} + (dv\sin\theta)^{-2}\right) \tag{5}$$

For stability to be achieved, CN ≈ 0.8 is the maximum (for $\tan\theta = 4$). The adjusted update scheme involves a lot of interpolation of missing values, which decrease the stability.

4.3.2 Perfectly Matched Layers

$$\frac{dp_\perp}{dt} + c^2\frac{do_\alpha}{d\alpha} + \kappa_\perp p_\perp = 0 \tag{6}$$

$$\frac{dp_\parallel}{dt} + c^2\frac{do_\gamma}{d\gamma} + \kappa_\parallel p_\parallel = 0 \tag{7}$$

The perfectly matched layers in the tilted grid become a bit more complex. This especially for the upper PML. The update equations for $p = p_\perp + p_\parallel$ are given by equations 6 and 7. Looking at the upper PML, so that the perpendicular direction is $\vec{e_y}$ and the parallel one is $\vec{e_x}$, these equations become:

$$\frac{dp_\perp}{dt} + c^2\frac{do_y}{dy} + \kappa_\perp p_\perp = 0,$$

$$\frac{dp_\parallel}{dt} + c^2\frac{do_x}{dx} + \kappa_\parallel p_\parallel = 0.$$

By defining $\vec{p} = p_\perp\vec{e_y} + p_\parallel\vec{e_x}$ and $\vec{\kappa} = \kappa_\perp\vec{e_y} + \kappa_\parallel\vec{e_x}$, these can be rewritten in one equation as follows

$$\frac{d}{dt}\vec{e_i}\cdot\vec{p} + c^2\vec{e_i}\cdot\nabla\vec{o}\cdot\vec{e_i} + |\vec{e_i}\cdot\vec{\kappa}|\,\vec{p}\cdot\vec{e_i} = 0$$

The absolute value of $\vec{e_i} \cdot \vec{\kappa}$ is important: if we chose $-\vec{e_y}$ instead of $\vec{e_y}$ as the perpendicular direction, the same equation should be gotten. Intuitively it is also non-sensical to have a negative damping-coefficient, as this would amplify. By now substituting $\vec{e_i} = \vec{e_w}$ or $\vec{e_i} = \vec{e_v}$, the equations for p_w and p_v are received.

It is important to note that this last step is not valid. Since this is an equation involving tensors, it cannot be generalized to any vector starting from two basis-vectors. This method, however, results in easy to implement PML's and, by visual validation, results in correct absorption.

4.3.3 Perfectly Hard Boundaries

The tilted boundary on the left side of the triangle is, as mentioned above, easy to implement. It just requires setting $o_w = 0$. The right-hand side isn't as simple to implement correctly (ie. such that only $o_n = 0$), instead a slightly incorrect alternative was used, setting all components of $\vec{o}$ to zero. Lastly the ground-plane had to be implemented. Neither of the simple approaches above are viable here, as $o_n = o_y$ is not stored directly so can't be set to zero, and neither can all components of $\vec{o}$ be set to zero. This last one is due to the fact that o_w and o_v aren't discretised at the same location, so that in reality the incoming wave will either see $o_w = 0$ or $o_v = 0$.

With this there is only one solution remaining, calculate o_y at the boundary by interpolation and remove this part from the stored o_v. In practice the opposite was done, o_x was calculated and o_v was set to only include this part. Approximating o_w at the ground by the one just above it, and using $o_x = o_w \sin\theta + o_v \cos\theta$, yields o_x. This boundary condition does not prevent waves from entering the PEC, which was mitigated by setting p equal to zero inside the PEC.

Finally another approach was tried. The goal is for o_y to be zero at the ground. Looking at its update equation,

$$\frac{do_y}{dt} + \frac{dp}{dy} = 0,$$

and using $o_y = 0$ for all t at $y = 0$, it follows that $\frac{dp}{dy} = 0$ at $y = 0$. As done before, this derivative can be written in terms of derivatives to u and v. The derivative towards v is easily approximated at the ground by using central difference, but towards u requires interpolation. From this equation p just below the ground can be set such that the boundary condition is achieved. Sadly this method turned out to be very unstable and it was not useable.

Boundaries with finite impedance were not studied for the tilted grid.

5 Conclusion

... The tilted grid makes the implementation of the tilted PEC easier and more accurate. But comes with a few drawbacks. Mainly it becomes less stable, leading to smaller timesteps and more phase-error. Besides this it makes the implementation of the PML and ground more complex. TODO: summarize results